

QUANTITATIVE APTITUDE CLOCK & CALENDER

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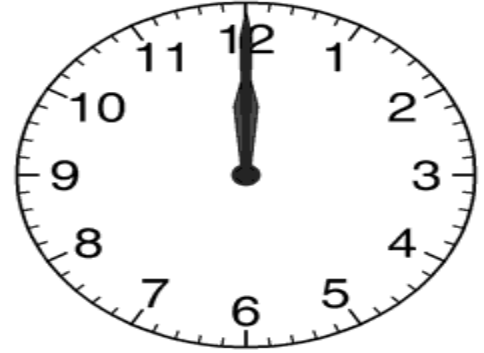


CLOCK

- ✓ Angle between needles
 - ✓ Specific needle position
 - ✓ Finding time for Specific needle position
 - ✓ Mirror Image and Water image of Clock
- 

CLOCK – Time Displayer

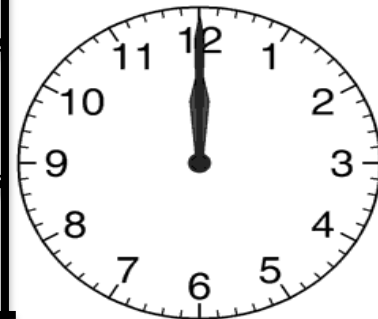
- ❖ Analog clocks have 12 major and 60 minor divisions, used to judge the time.
- ❖ 12 major divisions are numbered 1 to 12.
- ❖ There are 3 needles/hands
 - ✓ Hour Hand,
 - ✓ Minute Hand
 - ✓ Seconds Hand



CLOCK

Finding Angle between needles

	Hour Hand	Minute Hand	Seconds Hand
Time taken to cover 360°/complete rotation	12 hours	1 hour or 60 min	1 min or 60 sec
Degree covered in 1 hour or 60 minutes	$\frac{360^\circ}{12} = 30^\circ$	$\frac{360^\circ}{1} = 360^\circ$	$360^\circ \times 60$
Degree covered in 1 minute	$\frac{30^\circ}{60} = \left(\frac{1}{2}\right)^\circ$	$\frac{360^\circ}{60} = 6^\circ$	360°



****Every minutes there is a relative motion of $5\frac{1}{2}$ degrees, between the hour and the minute hand.**

Example

What is the angle between the minute hand and the hour hand when the clock strikes 20 past 2 (2:20)?

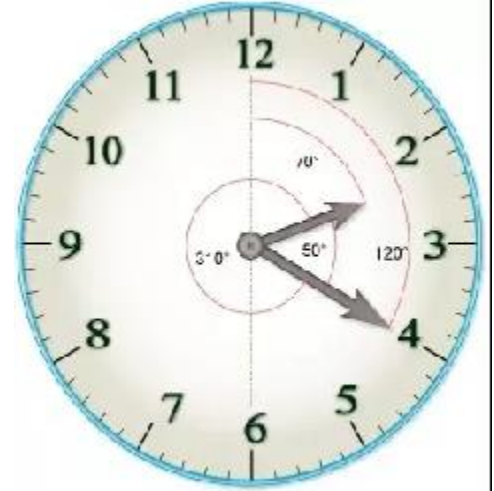
SHORT TRICK

Direct formula to determine the angle is,
Angle Between the Needles = $30H - \frac{11}{2} M$

Here $H = 2$ and $M = 20$

Therefore, the angle between the minute hand and the hour hand when the clock strikes 2:20

$$\begin{aligned} &= 30 \times 2^\circ - \frac{11}{2} \times 20^\circ \\ &= 60^\circ - 110^\circ \\ &= -50^\circ \end{aligned}$$



Example

What is the angle between the minute hand and the hour hand when the clock strikes 10 past 5 ?

Here $H = 5$ and $M = 10$

Therefore, the angle between the minute hand and the hour hand when the clock strikes 5:10

$$\begin{aligned} &= 30^\circ \times 5 - \frac{11}{2} \times 10^\circ \\ &= 150^\circ - 55^\circ \\ &= 95^\circ \end{aligned}$$

What is the angle between the minute hand and the hour hand when the clock shows quarter to 10?

Here $H = 9$ and $M = 45$

Therefore, the angle between the minute hand and the hour hand when the clock strikes 5:10

$$\begin{aligned} &= 30^\circ \times 9 - \frac{11}{2} \times 45^\circ \\ &= 270^\circ - 247.5^\circ \\ &= 22.5^\circ \end{aligned}$$

SPECIFIC NEEDLE POSITIONS

How many times in 12 hours, the needles make an angle of 0 degrees?

Every hour, the needles coincide once and thus makes a 0 degree angle.

But, between 11 to 1 this happens only once, hence in 12 hours this phenomenon occurs only 11 times.



12:00



1:05



2:11



3:16



4:22



5:27



6:33



7:38



8:44



9:49



10:55

How many times in 12 hours, the needles make an angle of 180 degrees?

Every hour, the needles make an angle of 180 degree once.

But, between 5 to 7 this happens only once, hence in 12 hours this phenomenon occurs only 11 times.

How many times in a day, the needles make a straight line?

Straight line includes 0 and 180 degrees both, thus in a day, it happens 44 times.

How many times in 12 hours, the needles make an angle of 90 degrees?

Every hour, the needles makes a 90 degree angle twice.

But, between 8 to 10 and between 2 to 4 this happens three times, hence in 12 hours this phenomenon occurs only 22 times (instead of 24).



SUMMARY FOR THE SPECIFIC NEEDLE POSITIONS

	12 Hours	In a Day (24 Hours)	EXCEPTIONS
0 Degree	11	22	At 12:00
180 Degree	11	22	At 6:00
Straight Line	22	44	At 12 & 6
90 Degree	22	44	At 3:00 & 9:00

FINDING TIME FOR SPECIFIC NEEDLE POSITIONS

CONCEPT

Find out the gap, that needs to be gain by min hand.

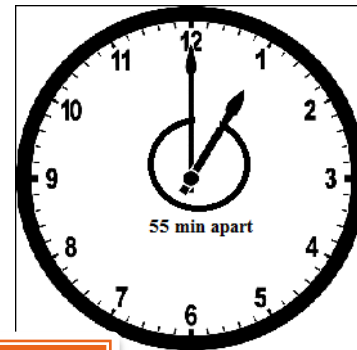
Multiply above obtained value by $60/55$.

At 12'O Clock min hand and hour hand coincide each other.

At 1'O Clock min hand and hour hand are 55 min apart.

So 55 min gap is created in 60 min.

So 1 min gap is created in $60/55$ min.



NOTE :-

Hands of the clock are in the same straight line and together (0°), they are 0 minute spaces apart.

Hands of the clock are in the same straight line but not together (180°), they are 30 minute spaces apart.

Hands of the clock are perpendicular to each other (90°), they are $15/45$ minute spaces apart.

At what time between 7 and 8 o'clock will the hands of a clock be in the same straight line but, not together?

Ans: When the hands of the clock are in the same straight line but not together, they are 30 minute spaces apart.

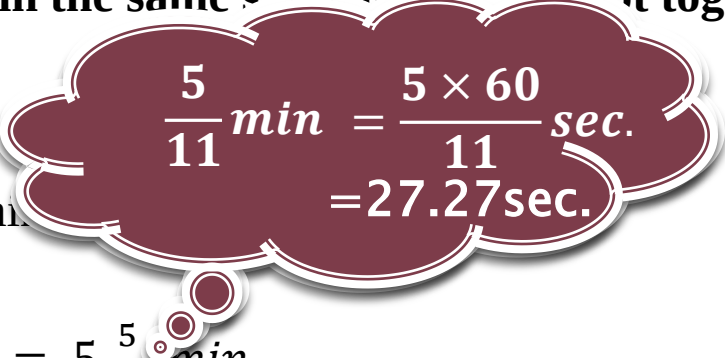
At 7 o'clock, they are 25 min. spaces apart.

Minute hand will have to gain only 5 min. spaces.

55 min. spaces are gained in 60 min.

5 min. spaces are gained in $\frac{60}{55} \times 5 \text{ min} = 5 \frac{5}{11} \text{ min.}$

So at 7:05:27 sec, hands of clock in same direction but not together.


$$\frac{5}{11} \text{ min} = \frac{5 \times 60}{11} \text{ sec.}$$
$$= 27.27 \text{ sec.}$$

At what time between 4 and 5 o'clock will the hands of a watch point in opposite directions?

Explanation:

At 4 o'clock, the hands of the watch are 20 min. spaces apart.

To be in opposite directions, they must be 30 min. spaces apart.

Minute hand will have to gain 50 min. spaces.

55 min. spaces are gained in 60 min.

50 min. spaces are gained in $\frac{60}{55} \times 50 \text{ min} = 54\frac{6}{11} \text{ min.}$

So at 4:54:32 sec, hands of clock are in opposite direction.

Q. At what time between 9 and 10 o'clock will the hands of a watch be together?

Explanation:

At 9 o'clock, the hands of the watch are 45 min. spaces apart.

To be together, min hand has to gain 45 min.

55 min. spaces are gained in 60 min.

45 min. spaces are gained in $\frac{60}{55} \times 45 \text{ min} = 49 \frac{1}{11} \text{ min.}$

So at 9:49:54 sec hands o clock in same direction but not together.

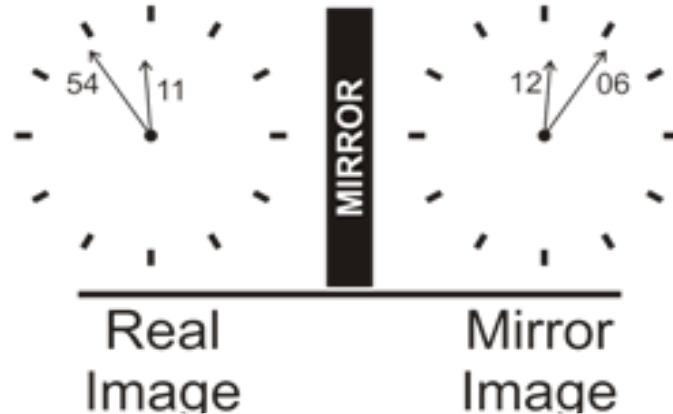


MIRROR IMAGE OF CLOCK

CONCEPT

Mirror gives the laterally inverted image of clock.

Lateral inversion means The right side of the item appears on the left side in such a picture, and vice versa.

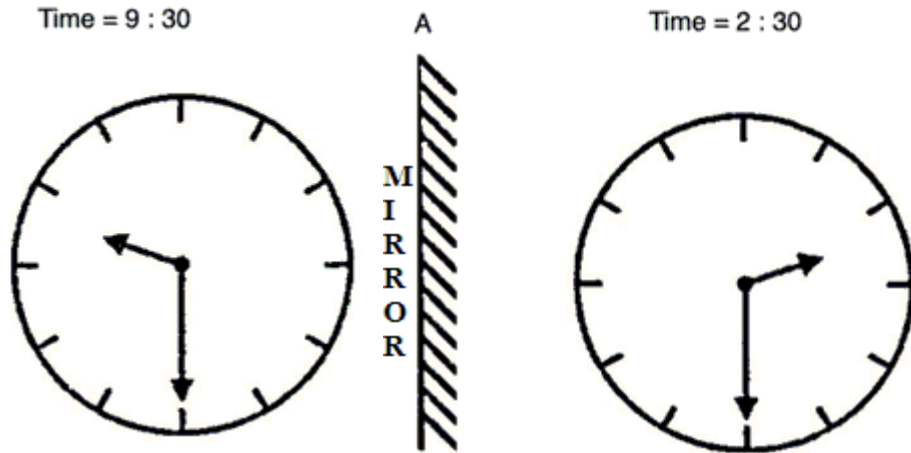


For each clock-based mirror image if clock has 12 hours, take the specified time and subtract it from **11:60** hours.

Looking into a mirror, the clock shows 9 : 30 as the time. The actual time is _____

- A. 2:30 B. 3:30 C. 3:30 D. 6:30

Answer: Option A



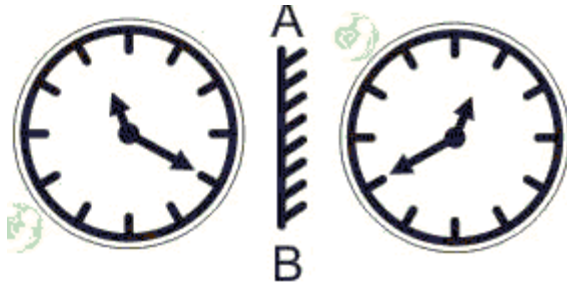
Answer:

$$11:60 - 9:30 \\ = 2:30$$

**When seen through a mirror, a clock shows 11 : 20.
The correct time is**

A. 11:40 B. 12:40 C. 12:20 D. 2:20

Answer: Option B



Answer:

11:60-11:20

=00:40

=12:40

Find the mirror image of a clock showing time as 6 : 30

A. 4:30 B. 5:30 C. 10:30 D. 12:230

Answer: Option B



Answer:

$$\begin{aligned} &11:60 - 6:30 \\ &= 5:30 \end{aligned}$$

WATER IMAGE OF CLOCK

CONCEPT

Water image is the vertically inverted image of clock.

Vertical inversion means The top side of the item appears as bottom side, and vice versa.



10 : 20



8:10

For every **water image** based on a clock,

If minutes given are **less than 30**, then subtract given time from **18:30 hrs.**

If minutes given are **more than 30**, then subtract given time from **17:90 hrs.**

If the water image of the clock shows time 6 hours 18 minutes, then what will be the actual time?

A. 11:15 B. 12:12 C. 11:18 D. 11:22

Answer: Option B

$$\begin{aligned} &18:30 - 6:18 \\ &= 12:12 \end{aligned}$$

Here given minutes are **less than 30**, so we subtract given time from **18:30 hrs.**

If the water image of the clock shows time 8:52, then what will be the actual time?

A. 9:38 B. 9:52 C. 10:38 D. 10:52

Answer: Option A

$$\begin{aligned} &17:90 - 8:52 \\ &= 9:38 \end{aligned}$$

Here given minutes are **more than 30**, so we subtract given time from **17:90 hrs.**

Q.2 What is the reflex angle between the hands of the clock at 3 o' clock ?
A. 90° B. 120° C. 180° D. 270°



Q.3 If a clock gain 5 minutes every hour and it is set right at 5 a.m., then what is the exact time when it shows 10 a.m. ?

A. 10 : 40 am B. 10 : 45 am C. 9 : 25 am D. 9 : 35 am

- 4. How many times in a day do the hands of a clock meet each other ?**
A. 20 times B. 22 times C. 28 times D. 44 times



5. At what time between 5 a.m. and 6 a.m. will the hands of the clock meet each other ?

A. 5: 30

B. 5: 25 : $\frac{5}{11}$

C. 5: 42: $\frac{5}{11}$

D. 5: 27: $\frac{3}{11}$

6. A clock, which gains 5 minutes every 12 hours, is set right at 5 a.m. What will be the real time, when the clock shows 5 p.m. On the following 5th day ?

A. 4: 55 p.m. B. 4: 05 p.m. C. 5: 05 p.m. D. 5: 55 p.m.



7. If the hands of a clock coincide every 64 minutes, then how much time does the watch gain per day ?

A. $360/11$

B. $365/11$

C. 90 minutes

D. NOT

8. At what times between 5: 30 and 6: 00 will the hands of the clock be at right angles ?

A. 43 min past 5

B. $40 \frac{7}{11}$ min past 5

C. $43 \frac{7}{11}$ min past 5

D. 45 min past 5



9. On planet FIAT, there are 10 days in a week. There are 36 hours in a day and each hour has 90 minutes while each minute has 60 seconds. As on earth, the hour hand covers the dial twice every day. Find the approximate angle between the hands of the clock on FIAT when the time is 14: 40 a.m.?
- A. 83 B. 74 C. 129 D. 65
- 

10. At 3:40, the hour hand and the minute hand of a clock form an angle of
- A. 120° B. 125° C. 130° D. 135°

12. The angle between the hour hand and the minute hand of a clock at 8:30 is
- A. 80° B. 75° C. 60° D. 105°

13. At what time between 4 and 5 o' clock will the hands of a clock be at right angle the first time?

A. 4:05

B. 4: 37 $\frac{5}{11}$ min

C. 4:05 $\frac{5}{11}$ min

D. 4: 37 $\frac{1}{11}$ min

14. A watch, which gains uniformly is 5 minutes slow at 5:00 a.m. on Monday and it is 5 minutes 30 seconds fast at 5:00 a.m. on the following Wednesday. When was the clock correct?

- A. 3:51 a.m. B. 4:41 a.m. C. 5.51 a.m. D. 3:45 a.m.**
- 

15. At what angle are the hands of the clocks inclined at 10 minutes to 9 ?
- A. 35.5 B. 34 C. 32.5 D. 35

17. At what time between 9 and 10 O' clock the hands of the clock are in a straight line but not together?

A. 57/11 min past 9

B. 180/11 min past 9

C. 184/11 min past 9

D. None of these

18. My watch gains 5 min every hour. How many degrees does the seconds hand move in every minutes?

A. 375

B. 380

C. 390

D. 365

19. At clock started at 10 am. By 20 minutes past 2 pm the hour hand has turned through how many degrees?

A. 120

B. 125

C. 130

D. 135

20. At what time between 5 and 6 O' clock, the hands of a clock are in the straight line and together?

- A. $229/11$ min B. $300/11$ min C. $149/6$ min D. None of these**
- 

CALENDER

- ✓ Utility of calendar
- ✓ Concept of ODD days
- ✓ Practice question

CALENDER UTILITY

To determine the day for the given combination of date, month and year.

CONCEPT

There are $365 \frac{1}{4}$ day in a year.

But to make the calculation easy, $\frac{1}{4}$ of a day from 3 continuous years is ignored

($\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = 1$) 1 complete day is added to every 4th year

which is then called as a LEAP year.

Thus there are 366 days in a leap year and 366th day is a day added to the month of February (29th Feb).

CONCEPT OF ODD DAYS

Number of extra days, outside a week is called as odd days.

In simple terms, if we divide the number of days by 7, the **remainder** gives the number of **ODD** days.

If we have to find out number of odd days in 48 $=48/7 \rightarrow 6$ odd days.

If we have to find out number of odd days in 14 $=14/7 \rightarrow 0$ odd days.

If we have to find out number of odd days in 106 $=106/7 \rightarrow 1$ odd days.

Number of odd days in an ordinary year

• $365/7$

---- 1 odd day

(1 is the remainder)

Number of odd days in an LEEP year

• $366/7$

---- 2 odd day

(2 is the remainder)

NUMBER OF ODD DAYS IN 100 YEARS

Dividing 100 by 4, we get 25 leap and 75 ordinary years.

But since 100th year (centuries other than a multiple of 400) is considered as ordinary year,

leap years = 24 and ordinary years = 76

Thus, number of odd days = $24 \times 2 + 76 \times 1$

$$= 48 + 76$$

$$= 124$$

$$= 124/7 \text{ ----- } 5 \text{ odd days.}$$

**In case of century year it
must be divisible by 400**

POINTS TO REMEMBER:

- ❖ **100** years have **5 odd** days.
- ❖ **200** years have **3 odd** days. (remainder of $10/7$) odd days.
- ❖ **300** years have **1 odd** days. (remainder of $15/7$) odd days.
- ❖ **400** years must have 20 odd days, but since 400th year is a leap year,
total number of odd days in 400 years = 0 (21/7 gives 0 remainder).
- ❖ So, 400, 800, 1200, 1600, 2000 years have 0 odd days.
- ❖ **Normal year** have **1 odd** day.
- ❖ **Leap year** have **2 odd** days.

Month	No of Days	No of ODD Days
January	31	3
February	28/29	0/1
March	31	3
April	30	2
May	31	3
June	30	2

Month	No of Days	No of ODD Days
July	31	3
August	31	3
September	30	2
October	31	3
November	30	2
December	31	3

❖ **Corresponding to the odd day count, the day of the week can be decided as per the given table.**

Days	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Odd Days	0	1	2	3	4	5	6

The last day of a century cannot be

- A. Monday B. Wednesday
C. Friday D. Saturday

Ans:- D. Saturday

Last day of century cannot be Tuesday, Thursday and Saturday.



Example

If 1st January, 2006 is Friday, which day will be on 1st January, 2009?

We will have to count the odd days, starting from 2006.

2006 – 1

2007 – 1 Thus, a total of 4 odd days,

2008 – 2

So we will add 4 to Friday and thus, it is Tuesday.

Example

If 21st February, 2014 is Friday, which day will be on 21st February, 2020?

We will have to count the odd days, starting from 21st February, 2014 .

21 st February, 2014 -21 st February, 2015	1 ODD day
21 st February, 2015 -21 st February, 2016	1 ODD day
21 st February, 2016 -21 st February, 2017	2 ODD day
21 st February, 2017 -21 st February, 2018	1 ODD day
21 st February, 2018 -21 st February, 2019	1 ODD day
21 st February, 2019 -21 st February, 2020	1 ODD day

Thus, a total of 7 odd days, =0 odd day

So this will be Friday on 21st Feb 2020

Example

If 30th December, 1996 is Monday, which day will be on 30th December, 1990?

We will have to count the odd days, starting from 1996 going backwards.

1996 – 2

1995 – 1

1994 – 1

1993 – 1

1992 – 2

1991 – 1

Thus, a total of 8 or 1 odd days,

So we will subtract 1 from Monday
and thus, it is Sunday.

Example

Today is Monday. After 61 days, it will be:

A. Tuesday B. Wednesday C. Thruway D. Staurday

$$\begin{aligned}\text{No of odd days in 61 days} &= \text{Remainder of } 61/7 \\ &= 5\end{aligned}$$

Thus, Monday + 5 days = Saturday

and thus, it will be Saturday.

Example

If April 11, 1911 was a Tuesday, what was the day on September 17, 1915?

A. Friday B. Thursday C. Sunday D. Tuesday

We will have to count the odd days, starting from 1911 to 1915.

Till April 11, 1912 – 2

Till April 11, 1913 – 1

Till April 11, 1914 – 1

Till April 11, 1915 – 1

Thus, Tuesday + 3 days = Friday

and thus, it will be Friday.

Remaining Days of April = 30 - 11 = 19 Odd Days

For May = 3 Odd Days

For June = 2 Odd Days

For July = 3 Odd Days

For August = 3 Odd Days

For September = 17 Days

Total Odd days = 3 odd Days

Example

Which day of the week was 26th July, 1999?

To get the day of the week, we need to find out the number of odd days for this combination of date, month and year.

Following steps are required:

STEP : 1

TARGET THE YEAR SECTION

Always subtract 1 from the given year

as 1999 is the running year in this case.

So we get $1999 - 1 = 1998$

Now $1998 = 1600 + 300 + 98$

0 odd + 1 odd + 98 odd + 24 odd

0 odd + 1 odd + 0 odd + 3 odd

→ 4 odd days

STEP : 2

TARGET THE MONTH SECTION

January – 3 Odd days

February – 0 Odd days

March – 3 Odd days

April – 2 Odd days

May – 3 Odd days

June – 2 Odd days

July – 26 Odd days

– 4 Odd days

total number

of odd days = $4 + 4 = 8$ odd days

1 odd days → Monday

Example

Which day of the week was 15th August, 1947?

To get the day of the week, we need to find out the number of odd days for this combination of date, month and year.

Following steps are required:

STEP : 1

TARGET THE YEAR SECTION

Always subtract 1 from the given year

as 1947 is the running year in this case.

So we get $1947 - 1 = 1946$

Now $1946 = 1600 + 300 + 46$

0 odd + 1 odd + 46 odd + 11 odd

→ 2 odd days

STEP : 2

TARGET THE MONTH SECTION

January – 3 Odd days

February – 0 Odd days

March – 3 Odd days

April – 2 Odd days

May – 3 Odd days

June – 2 Odd days

July – 3 Odd days

August – 15 Odd days

← 3 Odd days

total number

of odd days = $2 + 3 = 5$ odd days
→ Friday

- 1. If 15-March-2003 is Saturday, then find the day of the week on 23-January-2005?**
A. Sunday B. Monday C. Tuesday D. Wednesday



- 2. If 29-December-1972 is Friday then 22-April-1975 is on which day of the week?**
A. Sunday B. Monday C. Tuesday D. Wednesday



3. If 18-May-2001 is Friday, then 05-March 2005 is?

A. Sunday

B. Thursday

C. Friday

D. Saturday



4. A boy says his birthday is on 05-March. His brother's birthday is on 28-September. If his birthday comes on Monday, find the day of the week on his brother's birthday?

A. Sunday

B. Saturday

C. Friday

D. Thursday



5. Mary's daughter's birthday is on June-29, Sunday. Her husband's birthday is on x-mass day. Find the day of the week on her husband's birthday?
- A. Sunday B. Saturday C. Friday D. Thursday

- 6. If Raju's date of birth is 18-March-2002. How old is he by 17-April-2007?**
- | | |
|---------------------------|----------------------------------|
| A. 5 years 20 days | B. 5 years 29 days |
| C. 5 years 30 days | D. 5 years 1 month 2 days |



7. Vimala born on Monday, 26-April-1979. Find the day of week of her 24th birthday?

A. Sunday

B. Monday

C. Tuesday

D. Wednesday



8. Which of the following year is not a leap year?

A. 1948

B. 1984

C. 2074

D. 2096



- 9. If on 5th April 2015 is Sunday, then what day will become on 5th April 2016?**
A. Sunday B. Monday C. Tuesday D. Wednesday



- 10. If 26th March 2013 is Tuesday, then what day will become on 14th October 2013?**
- A. Sunday B. Monday C. Tuesday D. Wednesday**



11, Today is Wednesday, after 68 days, it will be

A. Friday

B. Thursday

C. Monday

D. Wednesday



12. The first republic day of India was celebrated on January 26, 1950. What day of the week was it?

A. Sunday

B. Monday

C. Wednesday

D. Thursday



13. How many days are there in x weeks x days?

A. $7x^2$

B. $7x$

C. $8x$

D. $8x+1$

- 14. On 8th Feb, 2005 it was Tuesday. What was the day of the week on 8th Feb, 2004?**
- A. Sunday B. Monday C. Tuesday D. Wednesday**



15. The calendar for the year 2007 will be the same for the year:
- A. 2014 B. 2016 C. 2017 D. 2018



16.

The last day of a century cannot be

A. Monday

B. Tuesday

C. Friday

D. Wednesday



- 17. The calendar for the year 2007 will be the same for the year:**
- A. 2014 B. 2016 C. 2017 D. 2018**



- 18. On 8th Dec, 2007 Saturday falls. What day of the week was it on 8th Dec, 2006?**
- A. Sunday B. Monday C. Thursday D. Friday**



- 19. If it was Tuesday on 3rd Jan, 2006. What was the day on Jan 3, 2010?**
- A. Sunday B. Monday C. Tuesday D. Wednesday**



20. If 13th March, 2005 was Wednesday, what was the day of the week on 13th March, 2004?

A. Sunday

B. Monday

C. Tuesday

D. Wednesday



THANK YOU

